

Context

A sales forecast is a prediction of future sales revenue based on historical data, industry trends, and the status of the current sales pipeline. Businesses use the sales forecast to estimate weekly, monthly, quarterly, and annual sales totals. It is extremely important for a company to make an accurate sales forecast as it adds value across an organization and helps the different verticals to chalk out their future course of actions. Forecasting helps an organization to plan its sales operations by region and provide valuable insights to the supply chain team regarding the procurement of goods and materials.

An accurate sales forecast process has many benefits which include improved decision making about the future and reduction of sales pipeline and forecast risks. Moreover, it helps to reduce the time spent in planning territory coverage and establish benchmarks that can be used to assess trends in the future.

Objective

SuperKart is an organization that owns a chain of supermarkets and food marts providing a wide range of products. They want to predict the future sales revenue of its different outlets so that they can strategize their sales operation across different tier cities and plan their inventory accordingly. To achieve this purpose, SuperKart has hired a data science firm, shared the sales records of its various outlets for the previous quarter, and asked the firm to come up with a suitable model to predict the total sales of the stores for the upcoming quarter.

Data Description

The data contains the different attributes of the various products and stores. The detailed data dictionary is given below.

- **Product_Id** - unique identifier of each product, each identifier having two letters at

the beginning followed by a number.

- Product_Weight - the weight of each product
- Product_Sugar_Content - sugar content of each product like low sugar, regular and no sugar
- Product_Allocated_Area - the ratio of the allocated display area of each product to the total display area of all the products in a store
- Product_Type - broad category for each product like meat, snack foods, hard drinks, dairy, canned, soft drinks, health and hygiene, baking goods, bread, breakfast, frozen foods, fruits and vegetables, household, seafood, starchy foods, others
- Product_MRP - maximum retail price of each product
- Store_Id - unique identifier of each store
- Store_Establishment_Year - the year in which the store was established
- Store_Size - the size of the store depending on sq. feet like high, medium, and low
- Store_Location_City_Type - the type of city in which the store is located like Tier 1, Tier 2, and Tier 3. Tier 1 consists of cities where the standard of living is comparatively higher than its Tier 2 and Tier 3 counterparts.
- Store_Type - the type of store depending on the products that are being sold there like Departmental Store, Supermarket Type 1, Supermarket Type 2, and Food Mart
- Product_Store_Sales_Total - total revenue generated by the sale of that particular product in that particular store

Guidelines

Full-code: The full code way is to write the solution code from scratch and only submit a final Jupyter notebook with all the insights and observations.

Best Practices for Full-code submissions

- The final notebook should be well-documented, with inline comments explaining the functionality of code and markdown cells containing comments on the observations and insights.
- The notebook should be run from start to finish in a sequential manner before submission.
- It is important to remove all warnings and errors before submission.
- The notebook should be submitted as an HTML file (.html) and NOT as a notebook